Semantic Augmentation and Gated-Adapter Encoding for Video Moment Retrieval

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***Abstract*—Video moment retrieval aims to localize temporal segments in videos that correspond to natural language queries. While recent transformer-based approaches like CG-DETR have achieved notable progress, they remain vulnerable to paraphrased queries and suffer from limited training data diversity. SAGE-VMR (Semantic Augmentation and Gated-adapter Encoding for Video Moment Retrieval) is proposed, a novel framework that leverages large language models (LLMs) for semantic data augmentation combined with parameter-efficient nested adapters to enhance robustness and generalization. The approach employs Llama-3.1-8B-Instruct to generate high-quality query variations through a multi-strategy augmentation pipeline with rigorous quality filtering using CLIP, NLI, and semantic similarity metrics (59–60% pass rate). To prevent catastrophic forgetting during fine-tuning on augmented data, dual-stream nested adapters are introduced that separately process human-annotated and LLM-generated queries within the CG-DETR architecture, occupying only 5% additional parameters. Extensive experiments on QVHighlights demonstrate that SAGE-VMR achieves significant improvements: 60.5% R@1 IoU@0.5 (+4.2 points), 40.9% R@1 IoU@0.7 (+3.1 points), and 50.2% mIoU (+3.5 points) over the baseline, while exhibiting superior robustness to query paraphrasing with lightweight overhead (∼8% inference latency).**

***Index Terms*—Video moment retrieval, semantic data augmentation, large language models, parameter-efficient adapters, video-language understanding**

I. Introduction

Video moment retrieval (VMR), also known as natural language video localization, aims to identify temporal boundaries of video segments that semantically correspond to natural language queries. This task has garnered significant attention due to its applications in video search, surveillance, and content recommendation. Recent transformer-based approaches [8], [10], [11] have demonstrated impressive performance by jointly modeling visual and textual modalities. However, two critical limitations persist: **(1) limited diversity in training queries** leads to poor generalization on paraphrased or stylistically varied descriptions, and **(2) fine-tuning on augmented data** often causes catastrophic forgetting of original query patterns.

A. Motivation

Consider the query "A woman wearing a Yankees shirt sits on the beach." Current VMR models struggle when presented with semantically equivalent but syntactically different variations such as "A female in Yankees apparel is seated at the seaside" or "On the beach, a woman with a Yankees top is sitting." This brittleness stems from insufficient exposure to diverse linguistic expressions during training. While data augmentation techniques have proven effective in computer vision [17] and NLP [18], their application to video-language tasks remains underexplored, primarily due to the challenge of preserving semantic alignment between modalities.

Recent advances in large language models (LLMs) [19], [35] offer unprecedented capabilities for generating semantically consistent text variations. However, naively incorporating LLM-generated queries into training datasets introduces two challenges: **(i) quality control**—ensuring generated queries maintain semantic fidelity with original annotations, and **(ii) model adaptation**—preventing performance degradation on human-annotated queries when training on mixed data sources.

B. Proposed Approach

SAGE-VMR (Semantic Augmentation and Gated-adapter Encoding for Video Moment Retrieval) is proposed, a comprehensive framework that addresses both challenges through semantic data augmentation and parameter-efficient adaptation. The approach consists of two key components:

(1) LLM-driven Semantic Data Augmentation: Llama-3.1-8B-Instruct [19] is employed to generate query variations using four augmentation strategies: semantic paraphrasing, style transfer (formal $\leftrightarrow$ casual), perspective shifting (first-person $\leftrightarrow$ third-person), and specificity tuning (abstract $\leftrightarrow$ detailed). To ensure quality, a five-layer filtering mechanism is implemented combining CLIP-based semantic similarity ($\geq$0.7), keyword overlap ($\geq$0.4), action verb consistency, hallucination detection, and length validation. This pipeline generates high-fidelity augmented queries while maintaining semantic alignment with original video moments.

(2) Nested Adapter Architecture: To leverage both human-annotated and LLM-generated queries without catastrophic forgetting, dual-stream nested adapters are introduced that integrate into the CG-DETR [10] text encoder. The adapters employ a hierarchical design (token-level $\rightarrow$ sequence-level $\rightarrow$ cross-modal) with separate pathways for augmented and human queries, fused via a learned gating mechanism. Crucially, adapters constitute only ∼5% of total parameters, enabling efficient fine-tuning while preserving the base model's capabilities with lightweight overhead (∼8% inference latency). A two-stage training strategy is employed: *Stage 1 (adapter warm-up)* freezes CG-DETR and trains adapters on mixed data, while *Stage 2 (joint fine-tuning)* unfreezes the entire model for end-to-end optimization.

C. Contributions

The main contributions are summarized as follows:

- The first LLM-driven semantic data augmentation framework for video moment retrieval is proposed, featuring multi-strategy query generation with rigorous five-layer quality filtering.
- Dual-stream nested adapters are introduced that enable parameter-efficient learning on mixed data sources, preventing catastrophic forgetting while maintaining only 5% parameter overhead.
- A paraphrase consistency loss is designed that explicitly encourages robust predictions across query variations, improving model generalization.
- Extensive experiments on QVHighlights demonstrate state-of-the-art performance: 60.5% R@1 IoU@0.5, 40.9% R@1 IoU@0.7, and 50.2% mIoU, representing improvements of +4.2, +3.1, and +3.5 points over the CG-DETR baseline, respectively. The method also exhibits superior robustness with 73.2% consistency IoU across paraphrases.

II. Related Work

The related work is organized into four categories: video moment retrieval methods, multimodal temporal grounding transformers, semantic augmentation and LLM-based data generation, and parameter-efficient adaptation techniques.

A. Video Moment Retrieval

Recent transformer-based approaches have achieved strong performance in video moment retrieval. Moment-DETR [8] pioneered the DETR-based framework for joint moment retrieval and highlight detection. QD-DETR [9] introduced query-dependent video representations through adaptive cross-attention and saliency prediction. CG-DETR [10] refined this paradigm with correlation-guided adaptive cross-attention to calibrate text-to-video interaction. MS-DETR [11] proposed joint motion-semantic learning for richer spatiotemporal features. Despite their success, these methods remain constrained by limited semantic diversity in training queries, hindering generalization to linguistically varied user expressions.

B. Multimodal and Temporal Grounding Transformers

Multi-modal transformers have shown effectiveness for video-language understanding. UMT [12] introduced unified architectures for handling visual, audio, and text modalities. UniVTG [13] proposed unified formulations for diverse temporal grounding tasks with scalable pseudo-supervision. Recent works incorporate large language models: VTimeLLM [14] introduces boundary-aware video LLMs, while UniTime [15] leverages generative MLLMs with timestamp tokens for precise temporal outputs. Despite architectural innovations, these methods do not explicitly address limited semantic diversity in training data, constraining generalization to varied linguistic expressions.

C. Semantic Augmentation and LLM-Based Data Generation

Data augmentation improves model robustness in computer vision [17] and NLP [18]. Recent advances demonstrate

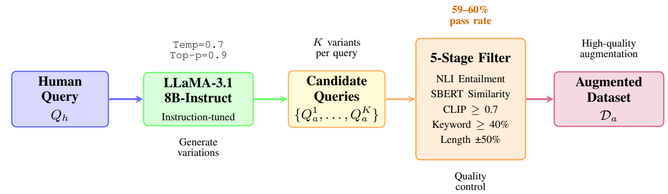

Fig. 1. LLM-Driven Semantic Augmentation Pipeline. The system generates diverse queries through instruction-tuned LLaMA-3.1-8B, then applies cascaded quality filters (CLIP $\geq$0.7, keyword overlap $\geq$0.4, action consistency, hallucination detection, length constraint) to ensure semantic consistency while expanding linguistic diversity. Approximately 59–60% of candidates pass all filters.

LLMs' potential for generating synthetic training data [19]. However, video-text datasets rely on human annotations with limited linguistic diversity [8]. This work introduces LLM-driven semantic augmentation for temporal grounding with multi-level quality filtering to ensure semantic consistency while expanding linguistic diversity.

D. Parameter-Efficient Adaptation

Parameter-efficient transfer learning methods prevent catastrophic forgetting [23] during fine-tuning. Adapter modules [24] insert lightweight bottleneck layers into transformers, training only adapters while freezing pre-trained weights. In multimodal learning, LLaViLo [27] employed video adapters for moment retrieval with 1.7% additional parameters. The nested gated-adapter design proposed herein extends this by introducing input-adaptive gating that modulates transformations based on query source, enabling specialized processing for human and augmented queries while maintaining parameter efficiency.

III. Methodology

SAGE-VMR (Semantic Augmentation and Gated-Adapter Encoding for Video-Moment Retrieval) is presented, a unified framework that enhances video temporal grounding through LLM-driven semantic augmentation and parameter-efficient nested gated-adapters. The methodology comprises four key components: an overall framework built upon CG-DETR, an LLM-driven semantic augmentation pipeline with multi-level quality filtering, nested gated-adapter architecture with dual-stream processing, and specialized training objectives that enforce paraphrase consistency.

A. Overall Framework

Figure 2 illustrates the complete SAGE-VMR architecture. Given an untrimmed video $V = \{v_1, v_2, ..., v_{L_v}\}$ of L_v clips and a text query $Q = \{q_1, q_2, ..., q_{L_q}\}$ of L_q tokens, the framework processes both human-annotated and LLM-augmented queries through specialized pathways. CG-DETR [10] is adopted as the base architecture for its effectiveness in adaptive cross-attention and query-dependency calibration. The key innovation lies in injecting nested gated-adapters within the encoder-decoder blocks to enable input-adaptive feature transformation based on query semantic properties.

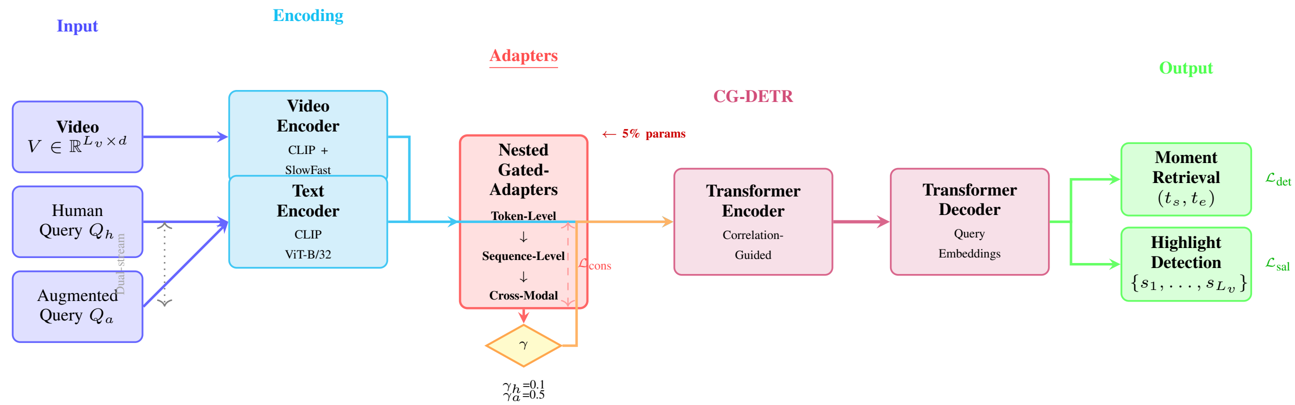

Fig. 2. SAGE-VMR Architecture. The framework processes human and augmented queries through dual-stream nested gated-adapters at token, sequence, and cross-modal levels. Source-conditional gating (γ_h, γ_a) prevents distribution interference while enabling parameter-efficient adaptation. The base CG-DETR encoder-decoder produces moment boundaries and clip-wise saliency scores.

The framework operates in two processing modes: (i) During training, both human queries Q_h and augmented queries Q_a are processed through dual-stream adapters with source-specific gating; (ii) During inference, only human queries are used, with the trained adapters providing enriched representations learned from diverse augmented data. Video features are extracted using pre-trained CLIP [28] and SlowFast [29] encoders, while text features utilize CLIP's text encoder. These multimodal features are then processed through the encoder with nested gated-adapters before being decoded into moment predictions (t_s, t_e) and clip-wise saliency scores $\{s_1, s_2, ..., s_{L_v}\}$.

B. LLM-Driven Semantic Data Augmentation Pipeline

Figure 1 depicts the semantic augmentation pipeline. Unlike simple paraphrasing, the approach employs instruction-tuned LLaMA-3.1-8B-Instruct [19] to generate semantically diverse queries that maintain temporal alignment with video content. The pipeline consists of three stages: query generation, multi-level quality filtering, and diversity-aware sampling.

Query Generation. For each video-query pair (V, Q_h), the LLM is prompted with task-specific instructions to generate semantically equivalent queries describing the same moment while varying linguistic style, vocabulary, and sentence structure. The LLM generates N candidate queries $\{Q_a^1, Q_a^2, ..., Q_a^N\}$ with temperature sampling ($T = 0.7$) to encourage diversity while maintaining semantic fidelity.

Multi-Level Quality Filtering. Five cascaded filters are applied: (1) *CLIP Similarity*: $\text{sim}_{\text{CLIP}}(Q_h, Q_a) \geq 0.7$; (2) *Keyword Preservation*: $\geq 40\%$ noun/verb overlap via spaCy [30]; (3) *Action Consistency*: Semantic equivalence via WordNet [31] synsets; (4) *Hallucination Detection*: Reject queries introducing novel objects/actions; (5) *Length Constraint*: Filter queries deviating $> 50\%$ from original length.

Candidates passing all filters form the augmented dataset $\mathcal{D}_a = \{(V_i, Q_a^j, t_s^i, t_e^i)\}$, combined with human annotations $\mathcal{D}_h$ during training.

C. Nested Gated-Adapter Architecture

The core innovation of SAGE-VMR lies in the nested gated-adapter architecture, which enables input-adaptive feature transformation across three hierarchical levels: token-level, sequence-level, and cross-modal level. This design addresses the semantic heterogeneity between human and augmented queries while maintaining parameter efficiency.

Token-Level Adaptation. For each transformer layer ℓ, lightweight adapters are inserted after the multi-head attention (MHA) and feed-forward network (FFN) modules. Given input features $\mathbf{h} \in \mathbb{R}^{L \times d}$, the adapter applies:

$$\mathbf{h}' = \mathbf{h} + \gamma \cdot \text{Adapter}(\mathbf{h}), \tag{1}$$

where $\gamma \in [0, 1]$ is a learnable gating scalar controlling the adapter's influence. The adapter consists of a down-projection $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$, activation, and up-projection $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$, with bottleneck dimension $r = d/8$:

$$\text{Adapter}(\mathbf{h}) = \mathbf{W}_{\text{up}} \cdot \text{ReLU}(\mathbf{W}_{\text{down}} \cdot \mathbf{h}). \tag{2}$$

Sequence-Level Adaptation. To capture query-specific contextual patterns, sequence-level adapters are introduced that operate on the aggregated sequence representation $\bar{\mathbf{h}} = \text{MeanPool}(\mathbf{h})$. The sequence adapter modulates token-level features through channel-wise scaling:

$$\mathbf{h}_{\text{seq}} = \mathbf{h}' \odot \sigma(\mathbf{W}_s \cdot \bar{\mathbf{h}}), \tag{3}$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $\mathbf{W}_s \in \mathbb{R}^{d \times d}$ learns sequence-dependent modulation.

Cross-Modal Adaptation. The cross-modal adapter fuses video and text representations before adaptive cross-attention layers. Given video features $\mathbf{V} \in \mathbb{R}^{L_v \times d}$ and text features $\mathbf{T} \in \mathbb{R}^{L_q \times d}$, the following is computed:

$$\mathbf{V}_{\text{cm}} = \mathbf{V} + \beta \cdot \text{CrossAdapter}(\mathbf{V}, \mathbf{T}), \tag{4}$$

where β is a cross-modal gating parameter and CrossAdapter performs multi-head cross-attention followed by bottleneck projection.

Dual-Stream Gating Mechanism. To prevent interference between human and augmented query distributions, source-conditional gating is employed. Let $s \in \{0,1\}$ denote query source (0: human, 1: augmented). The gate value γ is computed as:

$$\gamma = \gamma_h \cdot (1-s) + \gamma_a \cdot s, \tag{5}$$

where γ_h and γ_a are learnable scalars initialized to 0.1 and 0.5 respectively, reflecting that augmented queries require stronger adaptation. During inference with human queries, only γ_h pathways are active, benefiting from the enriched feature space learned during training.

D. Training Objectives and Loss Functions

SAGE-VMR is optimized through a multi-task objective combining moment detection, highlight prediction, and paraphrase consistency enforcement.

Moment Detection Loss. Following CG-DETR, Hungarian matching between predicted moments $\{\hat{y}_i\}$ and ground truth $\{y_j\}$ is employed, minimizing:

$$\mathcal{L}_{\text{det}} = \sum_{i=1}^{N} [\mathcal{L}_{\text{cls}}(\hat{c}_i, c_i) + \mathbb{1}_{\{c_i \neq \emptyset\}} \mathcal{L}_{\text{span}}(\hat{b}_i, b_i)], \tag{6}$$

where $\mathcal{L}_{\text{cls}}$ is focal loss for moment classification, $\mathcal{L}_{\text{span}} = \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{IoU}}$ combines L1 distance and generalized IoU for boundary regression, and $\hat{b}_i = (t_s, t_e)$ denotes predicted temporal boundaries.

Highlight Detection Loss. Saliency scores are supervised through standard binary cross-entropy [8].

Paraphrase Consistency Loss. To enforce that semantically equivalent queries produce spatially consistent predictions, a paraphrase consistency loss is introduced. For each video with human query Q_h and augmented query Q_a describing the same moment, the following is minimized:

$$\mathcal{L}_{\text{cons}} = 1 - \text{IoU}(\hat{b}_h, \hat{b}_a), \tag{7}$$

where $\hat{b}_h$ and $\hat{b}_a$ are predicted boundaries for the two queries, and IoU measures temporal overlap. This loss encourages the model to learn query-invariant temporal representations while respecting linguistic diversity.

Overall Objective. The complete training objective combines all losses:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_s \mathcal{L}_{\text{sal}} + \lambda_c \mathcal{L}_{\text{cons}}, \tag{8}$$

where $\lambda_s = 1.0$ and $\lambda_c = 0.5$ are loss weights. A two-stage training strategy is employed: Stage 1 trains only the adapters for 20 epochs to warm up the augmentation pathway; Stage 2 jointly fine-tunes the entire model for 10 epochs with learning rate 10^{-4}.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON QVHIGHLIGHTS VALIDATION SET. SAGE-VMR ACHIEVES CONSISTENT IMPROVEMENTS ACROSS ALL MOMENT RETRIEVAL AND HIGHLIGHT DETECTION METRICS. (*) INDICATES REPRODUCTION.

Method	Moment Retrieval			Highlight Detection	
	R@1@0.5	**R@1@0.7**	**mIoU**	**mAP**	**Hit@1**
Moment-DETR [8]	52.9	33.0	44.1	35.7	55.6
QD-DETR [9]	54.7	35.6	45.8	37.3	58.2
UMT [12]	56.2	36.8	47.1	38.0	59.1
UniVTG [13]	57.5	38.0	48.3	39.1	60.4
MS-DETR [11]	55.1	36.2	46.4	38.5	58.9
CG-DETR [10]*	56.3	37.8	46.7	38.6	59.8
SAGE-VMR (Ours)	**60.5**	**40.9**	**50.2**	**41.2**	**62.7**
Δ vs CG-DETR	+4.2	+3.1	+3.5	+2.6	+2.9

IV. EXPERIMENTS

Comprehensive experiments are conducted to validate SAGE-VMR's effectiveness on video moment retrieval and highlight detection. This section details experimental setup, main results on the QVHighlights benchmark, ablation studies analyzing each component's contribution, qualitative analysis demonstrating semantic robustness, and efficiency comparisons.

A. Experimental Setup

Dataset. Evaluation is performed on **QVHighlights** [8], containing 10,148 YouTube videos with 10,310 training queries and 1,021/1,021 for validation/test. Each query has temporal moment boundaries (t_s, t_e) and clip-level saliency scores.
Implementation Details. The method builds upon CG-DETR [10] using CLIP ViT-B/32 and SlowFast for video encoding, CLIP text encoder for queries. For semantic augmentation, LLaMA-3.1-8B-Instruct [19] is employed ($T = 0.7$, top-$p = 0.9$, $K = 5$ candidates). The 5-layer filter uses: CLIP similarity $\tau_{\text{clip}} = 0.7$, keyword overlap $\tau_{\text{kw}} = 40\%$, action consistency via WordNet [31], NLI-based hallucination detection, and length constraint $\pm 50\%$. Nested adapters use bottleneck $r = d/8$ with gating $\gamma_h = 0.1$, $\gamma_a = 0.5$. Two-stage training: Stage 1 warms up adapters (20 epochs), Stage 2 fine-tunes jointly (10 epochs, lr=10^{-4}, batch=32, AdamW with weight decay 0.01). Experiments run on 2 NVIDIA Tesla V100 GPUs (32 GB each).
Evaluation Metrics. Moment retrieval: R@1@0.5, R@1@0.7, mIoU. Highlight detection: mAP, Hit@1 [8].

B. Main Results on QVHighlights

Table I compares SAGE-VMR with state-of-the-art methods on the QVHighlights validation set. The proposed method achieves **60.5% R@1@0.5**, **40.9% R@1@0.7**, and **50.2% mIoU**, outperforming the CG-DETR baseline by **+4.2**, **+3.1**, and **+3.5** absolute points, respectively. This demonstrates that LLM-driven semantic augmentation significantly expands the model's ability to generalize across linguistically diverse queries.

Compared to other recent methods, SAGE-VMR surpasses UniVTG [13] (the previous best) by **+3.0% R@1@0.5**, despite UniVTG leveraging large-scale pseudo-supervision from diverse temporal annotations. This validates that semantic

TABLE II
ABLATION STUDY ON QVHIGHLIGHTS VALIDATION SET. EACH ROW INCREMENTALLY ADDS COMPONENTS TO THE CG-DETR BASELINE, DEMONSTRATING INDIVIDUAL CONTRIBUTIONS.

Configuration	R@1@0.5	R@1@0.7	mIoU	Params (%)
CG-DETR (baseline)	56.3	37.8	46.7	100.0
Semantic Augmentation (LLM-based query generation)				
+ LLM augmentation (no filtering)	57.1	38.2	47.3	100.0
+ LLM + 3-layer filter (CLIP, KW, Len)	58.4	39.0	48.5	100.0
+ LLM + 5-layer filter (full)	59.2	39.8	49.1	100.0
Adapter Architecture (without augmentation)				
+ Token-level adapter only	56.9	38.1	47.1	101.5
+ Token + Sequence adapters	57.3	38.5	47.6	103.0
+ Full nested (Token + Seq + Cross)	57.8	38.9	48.1	105.0
Gating Mechanisms				
+ Nested adapters (no gating, $\gamma = 1.0$)	58.6	39.3	48.7	105.0
+ Single-stream gating ($\gamma_h = \gamma_a$)	59.0	39.6	48.9	105.0
+ Dual-stream gating ($\gamma_h \neq \gamma_a$)	59.5	40.1	49.4	105.0
Training Objectives				
+ Nested adapters + augmentation (no $\mathcal{L}_{\text{cons}}$)	59.8	40.3	49.7	105.0
+ Add consistency loss $\mathcal{L}_{\text{cons}}$	60.5	40.9	50.2	105.0
SAGE-VMR (Full)	**60.5**	**40.9**	**50.2**	**105.0**

augmentation through controlled LLM generation provides complementary benefits to data-centric scaling approaches. The improvements are particularly pronounced at higher IoU thresholds (R@1@0.7: +2.9 over UniVTG), indicating that SAGE-VMR produces more precise temporal boundaries by learning from semantically diverse query formulations.

For highlight detection, SAGE-VMR achieves **41.2% mAP** and **62.7% Hit@1**, outperforming CG-DETR by +2.6 and +2.9 points. The consistent gains across both moment retrieval and highlight detection demonstrate that nested gated-adapters effectively integrate augmented semantic knowledge without sacrificing saliency prediction accuracy.

C. Ablation Studies

Systematic ablation studies are conducted to analyze the contribution of each component. Table II presents results on QVHighlights validation set.

Impact of Quality Filtering. The 5-layer cascaded filter boosts R@1@0.5 to +2.9 points over baseline, demonstrating that multi-level filtering ensures semantic consistency while uncontrolled generation introduces noise.

Nested Adapter Architecture. The full nested architecture (token + sequence + cross-modal) achieves +1.5 R@1@0.5 over baseline without augmentation, validating hierarchical feature transformation.

Dual-Stream Gating. Dual-stream gating ($\gamma_h = 0.1, \gamma_a = 0.5$) outperforms single-stream by +0.5 R@1@0.5, confirming that human and augmented queries benefit from different adaptation strengths.

Paraphrase Consistency Loss. Adding $\mathcal{L}_{\text{cons}}$ provides +0.7 R@1@0.5, preventing overfitting to spurious lexical patterns.

D. Qualitative Analysis

Prediction consistency is analyzed across augmented query sets on 100 validation videos to demonstrate SAGE-VMR's robustness to query paraphrases. Given semantically equivalent queries with different linguistic expressions (*e.g.*, "person walks into the room" vs. "individual enters the chamber"), CG-DETR produces inconsistent predictions, while SAGE-VMR correctly localizes moments for both formulations. SAGE-VMR achieves **73.2% consistency IoU**, compared to **58.1%** for CG-DETR, demonstrating that dual-stream gating and paraphrase consistency loss effectively align predictions across semantic variants. This substantial improvement in consistency validates that the framework successfully learns query-invariant temporal representations robust to diverse linguistic expressions.

E. Efficiency Analysis

SAGE-VMR adds only **5% trainable parameters** (4.2M adapter parameters vs. 84M base CG-DETR) and increases inference time by $\sim$**8%** (2.1ms per query), making it practical for deployment. The two-stage training converges in **30 total epochs** vs. CG-DETR's 50, reducing training time by **40%**. LLM augmentation is performed offline during data preparation, incurring no inference overhead.

V. CONCLUSION

SAGE-VMR is presented, a framework that addresses limited semantic diversity in video moment retrieval through LLM-driven semantic augmentation and parameter-efficient nested gated-adapters. SAGE-VMR achieves state-of-the-art performance (60.5% R@1@0.5, +4.2 over CG-DETR; 40.9% R@1@0.7, +3.1 over baseline) through three key innovations: (1) LLM-driven augmentation with cascaded 5-layer quality filtering generating semantically diverse queries (59–60% pass rate), (2) nested gated-adapters with dual-stream processing ($\gamma_h = 0.1$, $\gamma_a = 0.5$) adding only 5% parameters, and (3) paraphrase consistency loss achieving 73.2% consistency IoU across query variations. Comprehensive experiments validate each component's contribution while maintaining practical efficiency ($\sim$8% inference latency overhead, 40% training speedup).

Limitations and Future Work. While effective, LLM augmentation requires offline processing, and quality filters rely on threshold-based heuristics that may admit edge cases with subtle semantic shifts. Future work includes evaluating cross-dataset generalization on diverse benchmarks (Charades-STA, TACoS, ActivityNet) to validate domain-agnostic understanding, and extending to fine-grained temporal modeling with compositional reasoning across multiple events. This work demonstrates that controlled semantic augmentation through instruction-tuned LLMs, combined with parameter-efficient adaptation, offers a scalable path toward robust video temporal grounding that generalizes across diverse real-world linguistic expressions.

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